

The q -Binomial Foundation of the Bounded Fermionic Form Program: Proof of the Q -Transform and the Exactness of the Bottleneck (Round 2, Layer 3, Seed 7)

Loop Experiment

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Abstract

Fix a rank-3 profile c with $d = c_0 + c_1 + c_2 \not\equiv 0 \pmod{3}$ and let $H_{c,m} = (q; q)_m F_{c,m}$ be the normalized bounded generating function of cylindric partitions of profile c with maximum part at most m . We give the first complete proof of the Q -transform $Q_{n,c} = \sum_{m=0}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} H_{c,m}$ (previously stated by Seed 3 and verified numerically), prove its inverse $H_{c,m} = \sum_{n \leq m} \begin{bmatrix} m \\ n \end{bmatrix} Q_{n,c}$ unconditionally, and deduce: (i) the Bounded Fermionic Form Conjecture implies Warnaar's positivity conjecture with $Q_{n,c}$ equal to the fermionic coefficient a_n ; (ii) more sharply, the *existence* of a coefficientwise-nonnegative q -binomial expansion of the family $(H_{c,m})_m$ is *equivalent* to Warnaar's conjecture at c ; (iii) the entire bracket tower of Seed 8 has an unconditional expansion in the Q 's with manifestly nonnegative kernels, whence the MASTER conjecture is *equivalent* to Warnaar's conjecture, profile by profile and level by level. This makes the Layer-2 bottleneck exact and unique, and corrects the claimed equivalence chain “Monotonicity $\equiv f_0^{(m)} \geq 0 \equiv$ BFF first level”: the first two are strictly-weaker nonnegative-kernel projections of the conjecture, while the third is the conjecture itself.

1 Setup and conventions

Fix $r = 3$ and a profile $c = (c_0, c_1, c_2)$, $c_i \geq 0$, with $d = c_0 + c_1 + c_2$ and $\ell = \gcd(d, 3)$. Unless stated otherwise we assume $\ell = 1$, i.e. $d \not\equiv 0 \pmod{3}$ — exactly the regime of the Corteel–Dousse–Uncu/Warnaar positivity conjecture. All identities take place in the ring $\mathbb{Z}[[q]]$ of formal power series; *no polynomiality assumptions are needed* except in Remark 9.

Let $g_m \in \mathbb{Z}[[q]]$ denote the generating function of cylindric partitions of profile c with maximum part *exactly* m (so $g_0 = 1$), and set

$$F_{c,m} = \sum_{j=0}^m g_j, \quad F_c(z, q) = \sum_{m \geq 0} g_m z^m, \quad H_m := (q; q)_m F_{c,m}, \quad h_m := (q; q)_m g_m.$$

Thus $H_0 = 1$ and $h_m = H_m - (1 - q^m)H_{m-1}$ for $m \geq 1$. The bounded polynomials are

$$Q_n = Q_{n,c}(q) := (q; q)_n [z^n] \left((zq; q)_\infty F_c(z, q) \right).$$

We write $A \geq 0$ for a power series/polynomial with all coefficients nonnegative, and use both q -Pascal rules

$$\begin{bmatrix} m \\ j \end{bmatrix} = \begin{bmatrix} m-1 \\ j \end{bmatrix} + q^{m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix} = \begin{bmatrix} m-1 \\ j-1 \end{bmatrix} + q^j \begin{bmatrix} m-1 \\ j \end{bmatrix}. \quad (1)$$

The D -tower and bracket tower (Round 1; Seed 8, Layer 2) are

$$D_0^m = h_m, \quad D_{k+1}^m = D_k^m - q^{k+1} D_k^{m-1}; \quad f_{-1}^{(m)} = g_m, \quad f_k^{(m)} = (1 - q^{m-k}) f_{k-1}^{(m)} - q^{k+1} f_{k-1}^{(m-1)}.$$

We use two previously proved pieces of pure algebra (Seed 8, Layer 2, referee-checked): *Lemma T1*: $D_{k+1}^m = (q; q)_{m-k-1} f_k^{(m)}$ for $k \geq -1$, $m \geq k+1$; and its boundary case *Corollary T2*: $Q_n = f_{n-1}^{(n)} = D_n^n$. (Corollary T2 is also re-derived independently in Theorem 7 below.)

2 The truncated Euler lemma and the Q -transform

Lemma 1 (truncated Euler telescoping). *For every integer $K \geq 0$,*

$$\sum_{k=0}^K \frac{(-1)^k q^{\binom{k}{2}}}{(q; q)_k} = \frac{(-1)^K q^{\binom{K+1}{2}}}{(q; q)_K}.$$

Proof. Induction on K . For $K = 0$ both sides equal 1. Assume the identity at K ; adding the $(K+1)$ -st term,

$$\frac{(-1)^K q^{\binom{K+1}{2}}}{(q; q)_K} + \frac{(-1)^{K+1} q^{\binom{K+1}{2}}}{(q; q)_{K+1}} = (-1)^{K+1} q^{\binom{K+1}{2}} \frac{1 - (1 - q^{K+1})}{(q; q)_{K+1}} = \frac{(-1)^{K+1} q^{\binom{K+1}{2} + K+1}}{(q; q)_{K+1}},$$

and $\binom{K+1}{2} + K + 1 = \binom{K+2}{2}$. □

Theorem 2 (the Q -transform). *For every $n \geq 0$,*

$$Q_n = \sum_{m=0}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} H_m.$$

Proof. By Euler's expansion $(x; q)_\infty = \sum_{k \geq 0} (-1)^k q^{\binom{k}{2}} x^k / (q; q)_k$ at $x = zq$,

$$(zq; q)_\infty = \sum_{k \geq 0} \frac{(-1)^k q^{\binom{k+1}{2}}}{(q; q)_k} z^k.$$

Extracting the coefficient of z^n from the product with $F_c(z, q) = \sum_m g_m z^m$ gives

$$Q_n = (q; q)_n \sum_{m=0}^n \frac{(-1)^{n-m} q^{\binom{n-m+1}{2}}}{(q; q)_{n-m}} g_m. \quad (2)$$

On the other hand, substituting $H_m = (q; q)_m \sum_{j \leq m} g_j$ into the claimed right-hand side and using $\begin{bmatrix} n \\ m \end{bmatrix} (q; q)_m = (q; q)_n / (q; q)_{n-m}$,

$$\begin{aligned} \sum_{m=0}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} H_m &= (q; q)_n \sum_{j=0}^n g_j \sum_{m=j}^n \frac{(-1)^{n-m} q^{\binom{n-m}{2}}}{(q; q)_{n-m}} \\ &= (q; q)_n \sum_{j=0}^n g_j \sum_{k=0}^{n-j} \frac{(-1)^k q^{\binom{k}{2}}}{(q; q)_k} \quad (k = n - m) \\ &= (q; q)_n \sum_{j=0}^n g_j \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q; q)_{n-j}} \quad (\text{Lemma 1}) \end{aligned}$$

which is exactly (2). All sums are finite, so the interchange is legitimate in $\mathbb{Z}[[q]]$. $\square$

Remark 3. For $3 \mid d$ one has $Q_n = (q^3; q^3)_n [z^n]((zq; q)_\infty F_c)$; the Euler denominators $(q; q)_{n-m}$ then do not combine with $(q^3; q^3)$ -factorials into a single-base q -binomial, so no analogue of Theorem 2 in base q^3 results from this argument. This is outside the conjecture ($d \not\equiv 0 \pmod{3}$) and is not pursued.

3 Gauss inversion and the exactness of the bottleneck

Theorem 4 (Gauss inversion; classical). *Let $(a_n)_{n \geq 0}$, $(b_n)_{n \geq 0}$ be sequences in $\mathbb{Z}[[q]]$. The following are equivalent:*

$$(i) \quad b_n = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix} a_j \quad \text{for all } n; \quad (ii) \quad a_n = \sum_{m=0}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} b_m \quad \text{for all } n.$$

Moreover, for a given (b_n) the sequence (a_n) with property (i) is unique.

Proof. Both transforms are lower-unitriangular. It suffices to prove the orthogonality

$$\sum_{m=k}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} \begin{bmatrix} m \\ k \end{bmatrix} = \delta_{n,k}. \quad (3)$$

Using $\begin{bmatrix} n \\ m \end{bmatrix} \begin{bmatrix} m \\ k \end{bmatrix} = \begin{bmatrix} n \\ k \end{bmatrix} \begin{bmatrix} n-k \\ m-k \end{bmatrix}$ and substituting $i = n - m$, $J = n - k$, the left side equals

$$\begin{bmatrix} n \\ k \end{bmatrix} \sum_{i=0}^J (-1)^i q^{\binom{i}{2}} \begin{bmatrix} J \\ i \end{bmatrix} = \begin{bmatrix} n \\ k \end{bmatrix} (x; q)_J \Big|_{x=1} = \begin{bmatrix} n \\ k \end{bmatrix} \delta_{J,0},$$

by Gauss's finite q -binomial theorem $\sum_{i=0}^J (-1)^i q^{\binom{i}{2}} \begin{bmatrix} J \\ i \end{bmatrix} x^i = (x; q)_J$. Given (3), (i) $\Rightarrow$ (ii) and (ii) $\Rightarrow$ (i) follow by substitution and interchanging the (finite) sums; uniqueness holds because a unitriangular system over the commutative ring $\mathbb{Z}[[q]]$ has a unique solution. $\square$

Corollary 5 (inverse Q -transform; unconditional). *For every $m \geq 0$,*

$$H_m = \sum_{n=0}^m \begin{bmatrix} m \\ n \end{bmatrix} Q_n.$$

Equivalently, as generating functions, $\sum_{m \geq 0} H_m z^m = \sum_{n \geq 0} \frac{Q_n z^n}{(z; q)_{n+1}}$.

Proof. Theorem 2 is statement (ii) of Theorem 4 with $b_m = H_m$, $a_n = Q_n$; apply (ii) $\Rightarrow$ (i). The generating-function form follows from $\sum_{m \geq j} \begin{bmatrix} m \\ j \end{bmatrix} z^m = z^j / (z; q)_{j+1}$. $\square$

At $q = 1$: $H_m(1) = \sum_n \binom{m}{n} (K-1)^n = K^m$ with $K = \frac{1}{6}(d+1)(d+2)$, reconciling Welsh's evaluation $Q_n(1) = (K-1)^n$ with the orbit count $H_m(1) = K^m$ (Seed 2).

Corollary 6 (the bottleneck is exact and unique).

1. If $H_m = \sum_{j \leq m} \begin{bmatrix} m \\ j \end{bmatrix} a_j$ for all m , with $a_j \in \mathbb{Z}[[q]]$ independent of m , then $a_j = Q_j$ for all j . In particular the Bounded Fermionic Form Conjecture (such an expansion with a_j a manifestly nonnegative multisum) implies Warnaar's positivity conjecture at c , with the explicit formula $Q_n = a_n$.
2. Conversely, the family $(H_m)_m$ admits a coefficientwise-nonnegative q -binomial expansion if and only if $Q_n \geq 0$ for all n , i.e. if and only if Warnaar's conjecture holds at c .

Proof. (1) is the uniqueness in Theorem 4 plus Corollary 5. (2): the unique expansion coefficients are the Q_n , so a nonnegative expansion exists iff they are nonnegative. $\square$

Corollary 6(2) answers the characterization question in general: *any* sequence (b_m) in $\mathbb{Z}[[q]]$ has a unique q -binomial expansion, and it is nonnegative iff the inverse transform of (b_m) is nonnegative. There is no shortcut criterion to certify existence without exhibiting the coefficients: for the H -family, existence *is* the conjecture. (This is the precise sense in which Warnaar's open problem of bounded A_2 Andrews–Gordon identities and the positivity conjecture are two faces of one problem.)

4 Q -expansion of the towers and MASTER $\Leftrightarrow$ Conjecture

Theorem 7 (unconditional Q -expansions). *For all $0 \leq k \leq m$,*

$$D_k^m = \sum_{j=k}^m q^{(k+1)(m-j)} \begin{bmatrix} m-k \\ j-k \end{bmatrix} Q_j.$$

In particular $h_m = \sum_j q^{m-j} \begin{bmatrix} m \\ j \end{bmatrix} Q_j$, and $D_m^m = Q_m$. Moreover

$$\Delta_m := H_m - H_{m-1} = \sum_{j=1}^m q^{m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix} Q_j, \quad D_1^m = \Delta_m - q(1 - q^{m-1})\Delta_{m-1}.$$

Proof. For $k = 0$: by Corollary 5 and (1),

$$h_m = H_m - (1-q^m)H_{m-1} = \sum_j Q_j \left(\begin{bmatrix} m \\ j \end{bmatrix} - \begin{bmatrix} m-1 \\ j \end{bmatrix} + q^m \begin{bmatrix} m-1 \\ j \end{bmatrix} \right) = \sum_j Q_j q^{m-j} \left(\begin{bmatrix} m-1 \\ j-1 \end{bmatrix} + q^j \begin{bmatrix} m-1 \\ j \end{bmatrix} \right) = \sum_j Q_j$$

where the middle step used the first rule in (1) and the last used the second. For the inductive step,

$$D_{k+1}^m = \sum_j Q_j q^{(k+1)(m-j)} \left(\begin{bmatrix} m-k \\ j-k \end{bmatrix} - \begin{bmatrix} m-k-1 \\ j-k \end{bmatrix} \right) = \sum_j Q_j q^{(k+1)(m-j)} q^{(m-k)-(j-k)} \begin{bmatrix} m-k-1 \\ j-k-1 \end{bmatrix},$$

and $(k+1)(m-j) + (m-j) = (k+2)(m-j)$. The boundary $k = m$ gives $D_m^m = Q_m$, an independent re-derivation of Corollary T2. The Δ_m formula is the first difference of Corollary 5 via (1), and the last identity is immediate from $h_m = \Delta_m + q^m H_{m-1}$ and the definition of D_1^m . $\square$

Note that every kernel above is a monomial times a q -binomial coefficient: manifestly non-negative. This has the following consequence.

Theorem 8 (MASTER $\Leftrightarrow$ Conjecture, per profile). *Fix c with $d \not\equiv 0 \pmod{3}$ and $M \geq 0$. The following are equivalent:*

1. $Q_i \geq 0$ for all $i \leq M$;
2. for all $m \leq M$, all $-1 \leq k \leq m-1$ and all $0 \leq j \leq m-k-1$: $(q; q)_j f_k^{(m)} \geq 0$.

In particular, Warnaar's conjecture at c is equivalent to the full positivity part of the MASTER conjecture at c .

Proof. (1) $\Rightarrow$ (2). By Lemma T1, $f_k^{(m)} = D_{k+1}^m / (q; q)_{m-k-1}$ in $\mathbb{Z}[[q]]$ (the divisor has constant term 1, hence is invertible), so

$$(q; q)_j f_k^{(m)} = \frac{D_{k+1}^m}{(q^{j+1}; q)_{m-k-1-j}}.$$

By Theorem 7, $D_{k+1}^m = \sum_{i \geq k+1} q^{(k+2)(m-i)} \begin{bmatrix} m-k-1 \\ i-k-1 \end{bmatrix} Q_i \geq 0$ under (1), and $1/(q^{j+1}; q)_{m-k-1-j} = \prod_{s=j+1}^{m-k-1} (1-q^s)^{-1}$ is a nonnegative power series. A product of nonnegative series is nonnegative. (The row $k = -1$ reads $(q; q)_j g_m = h_m / (q^{j+1}; q)_{m-j} \geq 0$.)

(2) $\Rightarrow$ (1). The cell $(k, j) = (m-1, 0)$ is $f_{m-1}^{(m)} = Q_m \geq 0$ (Corollary T2), for each $m \leq M$. $\square$

Remark 9 (exactness of MASTER is automatic). *Suppose in addition that $H_m \in \mathbb{Z}[q]$ (Seed 3's polynomiality theorem, $\ell = 1$), so all $D_{k+1}^m \in \mathbb{Z}[q]$. If the conjecture holds at c and $f_k^{(m)} \neq 0$, then $(q; q)_{m-k} f_k^{(m)} = (1 - q^{m-k}) D_{k+1}^m$ has strictly negative leading coefficient, since the nonzero nonnegative polynomial D_{k+1}^m has positive leading coefficient. Hence the MASTER range $j \leq m-k-1$ is sharp wherever $f_k^{(m)} \neq 0$: the conjectured “fails at $j = m-k$ ” clause needs no separate proof.*

5 The corrected equivalence chain

Corollary 10 (implication structure of the Layer-2 conjectures). *Fix c with $d \not\equiv 0 \pmod{3}$. Then, as statements about all m, n simultaneously:*

$$\underbrace{\text{nonneg. } q\text{-binomial expansion of } (H_m)}_{\text{BFF, first level}} \iff \underbrace{Q_n \geq 0 \ \forall n}_{\text{Warnaar}} \iff \text{MASTER}$$

and each of these implies (via the nonnegative kernels of Theorem 7):

$$\text{Monotonicity } (\Delta_m \geq 0 \ \forall m), \quad h_m \geq 0 \ \forall m, \quad f_0^{(m)} \geq 0 \ \forall m, \quad D_k^m \geq 0 \ \forall k, m.$$

Monotonicity and $f_0^{(m)} \geq 0$ are inequivalent-looking projections ($D_1^m = \Delta_m - q(1 - q^{m-1})\Delta_{m-1}$; positivity does not transfer through the mixed-sign factor $(q; q)_{m-1}$ of Lemma T1 in either direction), and no implication from any of these weaker statements back to the conjecture is known (cf. BA24).

Proof. The equivalences are Corollary 6(2) and Theorem 8. The one-way arrows: each listed quantity is, by Theorem 7 (and Theorem 8(a) for the f -cells), the image of the vector $(Q_j)_j$ under a componentwise-nonnegative kernel. $\square$

Structurally: *every* bounded positivity statement on the Layer-2 board is a nonnegative unitriangular kernel applied to the single vector $(Q_n)_n$, and the conjecture is the statement that the source vector itself is nonnegative — the unique maximal element of this family. The strategic consequence for Layer 3+ is that no finite collection of kernel-image positivities (Monotonicity, $h_m \geq 0$, $N_n \geq 0$, individual MASTER cells) can suffice; one must either produce the expansion coefficients positively (the BFF route: explicit bounded A_2 Andrews–Gordon multisums, where all m -dependence sits in a single $\begin{bmatrix} m \\ n \end{bmatrix}$), or prove positivity of Q_n by a mechanism outside this transform family (e.g. the Corteel–Welsh/R-relation or Φ_3 -division routes).

6 Worked example and verification

$d = 2$, profile orbit $(1, 1, 0)$: the proved Rogers–Ramanujan form $H_m = \sum_j q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}$ (Seeds 3/4) is a q -binomial expansion with $a_j = q^{j^2}$; Corollary 6(1) yields instantly $Q_n = q^{n^2}$, recovering the Layer-2 theorem of Seeds 1/6. Conversely $H_m = \sum_n \begin{bmatrix} m \\ n \end{bmatrix} q^{n^2}$ illustrates Corollary 5.

All results were verified exactly in $\mathbb{Z}[q]$ (script `scratch/scripts/seed7_R2L3_verify.sage`): Lemma 1 for $K \leq 12$; orthogonality (3) for $n \leq 8$; Theorems 2, 7 and Corollary 5 for $d = 4$ (15 profiles, $n \leq 6$), $d = 5$ (21 profiles, $n \leq 5$), $d = 7$ (36 profiles, $n \leq 4$); the MASTER cells and the exactness remark in the same ranges. The computation pipeline itself (EMD/ H -recursion) was validated for the first time against brute-force enumeration of cylindric partitions ($m \leq 3$, coefficients to q^{10}); this revealed that the project-standard EMD convention labels profiles in the *reversed* orientation relative to the interlacing definition in `conjecture.tex` (brute-force profile $(0, 3, 1)$ matches the H -recursion on the orbit of $(0, 1, 3)$, etc.) — a global relabeling harmless to all per-profile theorems, but essential bookkeeping when matching fermionic forms against Warnaar’s tables (cf. conflict C2).